

- 36** Two of the requirements of an efficient gas exchange system are a large surface area and a short diffusion distance.

Which row correctly describes how alveoli are adapted to meet these requirements?

|          | large surface area                                        | short diffusion distance                                                   |
|----------|-----------------------------------------------------------|----------------------------------------------------------------------------|
| <b>A</b> | elastin fibres prevent the alveolus wall from collapsing  | an extracellular layer inside the alveolus wall contains blood capillaries |
| <b>B</b> | gases dissolve in a layer of liquid to speed up diffusion | alveolar walls are next to capillaries                                     |
| <b>C</b> | alveoli are folded and interconnected                     | walls of alveoli are only one cell thick                                   |
| <b>D</b> | walls of alveoli are formed of squamous epithelial cells  | red blood cells are very close to capillary walls                          |